

Homework

Assigning Instrumental Variables

Case of Bulgarian Communist Era Wages (1986)

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Class Model

Specification

Dependent Variable: LOG(WAGE)

Method: Two-Stage Least Squares

Date: 03/19/26

Sample: 1 10500 IF WAGE > 0

Included Observations: 10331

Instrument Specification: C, GRADES, FL, READ, RELIG

Class Model

Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.372735	0.156603	2.380133	0.0173
EDU	-0.033558	0.023950	-1.401175	0.1612
INNOV	0.011635	0.022960	0.506721	0.6124
MAJOR	0.530028	0.020033	26.45774	0.0000
MALE	0.238713	0.019011	12.55664	0.0000
BKP	0.404941	0.197271	2.052710	0.0401
AGE	0.099950	0.003029	32.99719	0.0000
AGE^2	-0.001028	2.69E-05	-38.18487	0.0000
R-squared	0.307434	Mean dependent variable		2.667893
Adjusted R-squared	0.306964	S.D. dependent var		0.829038
S.E. of regression	0.690164	Sum squared residuals		4917.112
F-statistic	696.8929	Durbin-Watson statistic		1.959189
Prob(F-statistic)	0.000000	Second-Stage SSR		4776.209
J-statistic	147.8605	Instrument rank		10
Prob(J-statistic)	0.000000			

Class Model

Exogeneity Test

Equation: CLASS_EQ Workfile: COMMUNISM BCP::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Endogeneity Test
 Equation: CLASS_EQ
 Endogenous variables to treat as exogenous: EDU BKP
 Specification: LOG(WAGE) C EDU INNOV MAJOR MALE BKP AGE AGE^2
 Instrument specification: C GRADES FL INNOV MAJOR MALE READ
 RELIG AGE AGE^2
 Null hypothesis: EDU BKP are exogenous

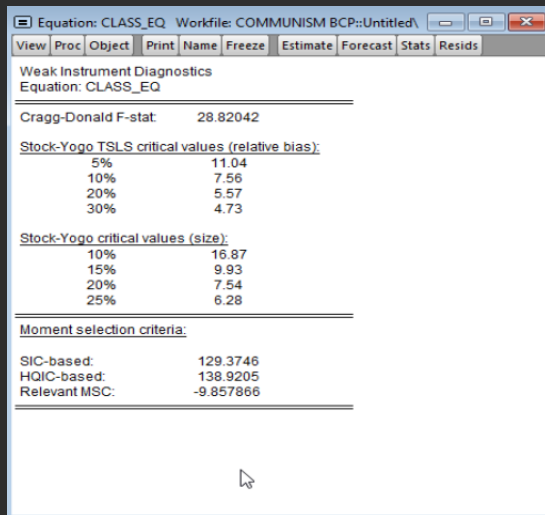
	Value	df	Probability
Difference in J-stats	156.6681	2	0.0000

J-statistic summary:

	Value
Restricted J-statistic	312.4926
Unrestricted J-statistic	155.8245

Class Model

Stock Yogo



Best Model

Specification

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Instrument Specification: C, GRADES, GRADES² FL, FL²
INNOV, INNOV², RELIG

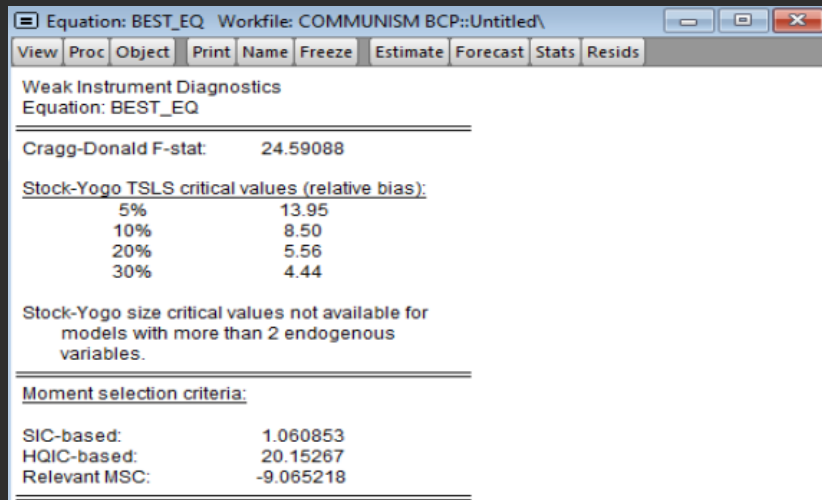
Best Model

Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.750072	0.143508	-5.226673	0.0000
EDU	0.667379	0.068620	9.725742	0.0000
EDU ²	-0.103346	0.009839	-10.50400	0.0000
AGE	0.098050	0.002982	32.88007	0.0000
AGE ²	-0.000998	2.77E-05	-35.99323	0.0000
BKP	0.448531	0.170358	2.632866	0.0085
MAJOR	0.725095	0.026382	27.48472	0.0000
MALE	0.183775	0.020093	9.146081	0.0000
READ	0.148789	0.029493	5.044949	0.0000
R-squared	0.236575	Mean dependent var	2.667893	
Adjusted R-squared	0.235983	S.D. dependent var	0.829038	
S.E. of regression	0.724646	Sum squared resid	5420.201	
F-statistic	574.4696	Durbin-Watson stat	1.953576	
Prob(F-statistic)	0.000000	Second-Stage SSR	4686.561	
J-statistic	38.03247	Instrument rank	13	
Prob(J-statistic)	0.000000			

Best Model

Stock Yogo



Ending remarks

I am delighted for your attention.

Q & A?